

Lab 8

Evaporation and Clouds

Show me my silver lining

Equipment List

- Ruler
- Hygrometer
- Digital Thermometer
- Ulanzi light / flashlight
- 250 mL beaker (2)
- 500 mL beaker
- Cloud chamber kit (cube with tube, felt, alcohol, caulk, baking tray, black vinyl, uranium rock)
- Tall cloud jar w/ metal bowl for lid
- Short cloud jar w/ glass bowl for lid
- Dry ice chunk (instructor-provided)
- Regular ice (in cooler)

8.1 Introduction and Definitions

8.1.1 Atmospheric pressure and a warm-up experiment

We have read that the the average atmospheric pressure at sea level on the Earth is about one kilogram force-equivalent per square centimeter of area ($2.2 \text{ lb/cm}^2 = 14.2 \text{ lb/in}^2 = 14.2 \text{ psi} = 101 \text{ kPa}$). This pressure is literally the weight of the column of air above us. Because liquid water is so much denser than air one can achieve an additional 1 atm of pressure by submerging below just 10 m of water¹. Air pressure drops off exponentially with height in the atmosphere with a *scale height* (distance over which it drops by a factor of $\frac{1}{e} \approx \frac{1}{3}$) of 8 km, going to zero pressure — a so-called *vacuum* — in space. The air pressure at Mount Everest base camp, 5,400 m above sea level, is about half what it is at sea level: $\sim 50 \text{ kPa}$.

Mini-experiment: We do not “feel” the air pressure because it has the same value outside of our bodies (the air molecules beating against our skin) as it does internally (the water and blood molecules below our skin). But there is a simple experiment we can do to observe its effects:

¹The pressure increase from the surface of a fluid to a depth h is given by the equation $|\Delta P| = \rho gh$, where ρ is the density of the fluid (1000 kg/m^3 for water) and $g \approx 10 \text{ m/s}^2$ is the acceleration due to gravity.

evacuate the air in a soda can. Together with your table of four students, put a tiny amount of water into an empty soda can. Heat this on a hot plate until the water is boiling and, most importantly, the can is filled with water vapor². By filling the can with pure water vapor, most of the air has been pushed out of the can. If we could then rapidly condense the water vapor into liquid we would have an approximate vacuum inside the can (because the volume of *liquid* water associated to a can full of water *vapor* is miniscule). Do this with your group by gently but quickly removing the can from the hotplate with tongs, and then turning it upside down in a bowl of cold water. Observe any noticeable effect on the can and document the experiment here.

8.1.2 Relative humidity, absolute humidity, partial pressure of water

Last week in lab we boiled water in a 250 mL beaker. We observed that, as expected for our classroom conditions at atmospheric pressure, boiling occurred when the water reached 100°C. Look back at the phase transition graph that you marked up last week, pressure vs. temperature, and remind yourself why this was expected, i.e., following the horizontal line of $p = 1$ atm to the **coexistence curve** between liquid and vapor water. You should have noticed in this experiment, however, that there was significant mass loss of liquid water prior to the “boiling point.” Mass of water dropped off rapidly once the water temperature reached 100°C (in some experiments at a rate of about 3.5×10^{-2} g/s), but there was a measurable mass decline when the temperature was in the range 50–100°C as well (a rate of about 1.0×10^{-2} g/s). The reason for this behavior is because:

- (i) The phase diagram we used last week pertains to *pure water*, i.e., liquid water in a closed container with only water vapor and ice. This interpretation is good for water in the bulk of the fluid, below its surface. Once the internal liquid water molecules reach 100°C the intermolecular forces weaken and they are likely to separate from their neighboring molecules to form water vapor instead, which then bubbles to the surface (“boiling”).
- (ii) Water molecules at the surface, however, are exposed to air, so they are *not* in a “pure” liquid-vapor water situation. Generally air contains only some small percentage (mole fraction) of water vapor. It turns out that³ the same phase transition graph can be used to determine the coexistence curve, but the pressure value one should use is the so-called

²As we will “see” today, actual water vapor is invisible, so we only see its presence in the moments that it comes in contact with colder air and condenses into tiny liquid drops.

³The physical reason for this result is due to the ideal gas law, where, as long as the densities are quite low, all types of gas molecules treated as the identical non-interacting particles; so the liquid water at the surface only “sees” a low-density water-gas within the total air gas.

partial pressure of water in air:

$$e = f \times p_{\text{atm}} = \frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{\text{dry air}}} \times p_{\text{atm}}$$

where e is the variable⁴ generally used for partial pressure, and n is the variable for the number of moles, so f is the *mole fraction of water in air*. Because the mole fraction of water in air is generally significantly less than one (on the order of 1%), the partial pressure of water in air is significantly less than atmospheric pressure.

So, your boiling experiment was actually two separate experiments occurring at once: a pure-water phase transition occurring below the surface, and a liquid-to-vapor-in-air transition occurring at the surface. Make a sketch of your boiling water beaker and use arrows to label the two different water vapor pressures at the two different locations. Give these pressures their variable names and estimate their values roughly (using the “1%” estimate given above).

We will do this more precisely in a moment, but quickly return to your phase diagram from last week and estimate a vaporization temperature for the water at the surface, where $e \approx 0.01 \times p_{\text{atm}}$. State your process and record an approximate temperature value (as best you can read off that graph) here.

The vaporization temperature for liquid water in contact with air is called the **dew point temperature**. Let's determine this value more carefully. First look up the “current conditions” on a weather page (e.g., search for “NWS 7-day grand rapids”) and record the current temperature, the **relative humidity** (expressed as a percentage), and the dew point temperature here, [Note: If it is raining and/or the humidity is near 100%, choose another data point within the “3 Day History” when the humidity was lower.]

⁴This is not the same “e” mentioned above in the “scale height” discussion; that was the exponential constant, or *Euler's number*, $e = 2.718 \dots$

Now examine the so-called **psychrometric chart** shown in Figure 8.1. Use your temperature (horizontal axis) and relative humidity (curved lines labeled at top) values to determine the associated **absolute humidity** (vertical axis), which is expressed here as a ratio of water mass to (dry) air mass:

$$AH = \frac{m_{\text{H}_2\text{O}}}{m_{\text{dry air}}}$$

To determine the partial pressure of water in air, we must find the mole fraction, f , from this quantity. Let's derive the relationship between AH and f now. (i) Recall that the number of moles of a gas can be expressed as $n = m/M$ where m is the actual mass of the volume of gas and M is the **molar mass** of the gas (read off the periodic table). Find M for both water and for dry air. Remember that air is basically 80% nitrogen (N_2) and 20% oxygen (O_2), so $M_{\text{air}} = 0.8M_{\text{N}_2} + 0.2M_{\text{O}_2}$.

(ii) Now write out the *mole ratio* $n_{\text{H}_2\text{O}}/n_{\text{air}}$ and simplify it so that it is expressed in terms of AH.

(iii) Write out the equation for the mole fraction of water in air, f , given implicitly in the equation for partial pressure above. Using the mole ratio, you should be able to re-write this in terms of the AH value and some numerical value that comes from the molar mass values calculated above.

Check your answers with your table and your instructor.

Now convert the AH value you found from the psychrometric chart into a mole fraction value, and, finally, use that to calculate a partial pressure value, e , in atmospheres (atm).

Finally, using the new phase diagram provided in Figure 8.1, determine the vaporization temperature of water in air, given the partial pressure you determined. Compare this to the dew point value in the weather report.

You should find in the weather data, and in your calculated verification, that the dew point temperature is well below the air temperature. What do you think are the implications of this fact? What will happen if you spill water on the table? What is the meaning of “relative humidity,” and what happens to the relative humidity as the air temperature decreases to the dew point? How does the “distance to the dew point” (and the relative humidity) affect your body’s ability to cool by perspiration?

8.2 Experiments and Analysis

Safety Notes:

- For the “cloud chamber” experiment, your instructor will provide you with a chunk of solid carbon dioxide (“dry ice”). This is about -80°C and it will burn you if you touch it. So, please **use the thermal gloves to handle the dry ice** if you need to (but you probably won’t need to handle it).
- Also for the cloud chambers, you will be provided with a small radioactive rock (containing uranium and other things). As we will see, this emits “alpha particles” (helium nuclei). These particles are not able to penetrate your skin and are therefore safe. However, you would not want to consume these rocks as the alpha particles could damage you internally. So do not eat the radioactive rock, and also **do not touch the radioactive rock without nitrile gloves**. If you do happen to touch the rock, then wash your hands carefully.

8.2.1 Observing the dew point and frost point

We will not be performing our experiments outside, so we need to calculate the partial pressure of water for the conditions in this room. Measure the temperature of the room with the IR thermometer gun (look at a bunch of objects that should be in thermal equilibrium with the air), and the relative humidity using the hygrometer. Also give uncertainties for these measurements (and your rationale for these values).

Use the process from the introduction to convert your relative humidity value to an absolute humidity (AH) and finally to a partial pressure, e , in atmospheres. Then use the coexistence curve in the phase diagram to determine the vaporization temperature (the dew point) in this room. Record the value and any intermediate calculations here.

Put some “cold” tap water in a 250 mL beaker and measure its temperature. Then slowly add ice to the water to reduce its temperature. Observe what happens to the outer walls of the beaker.

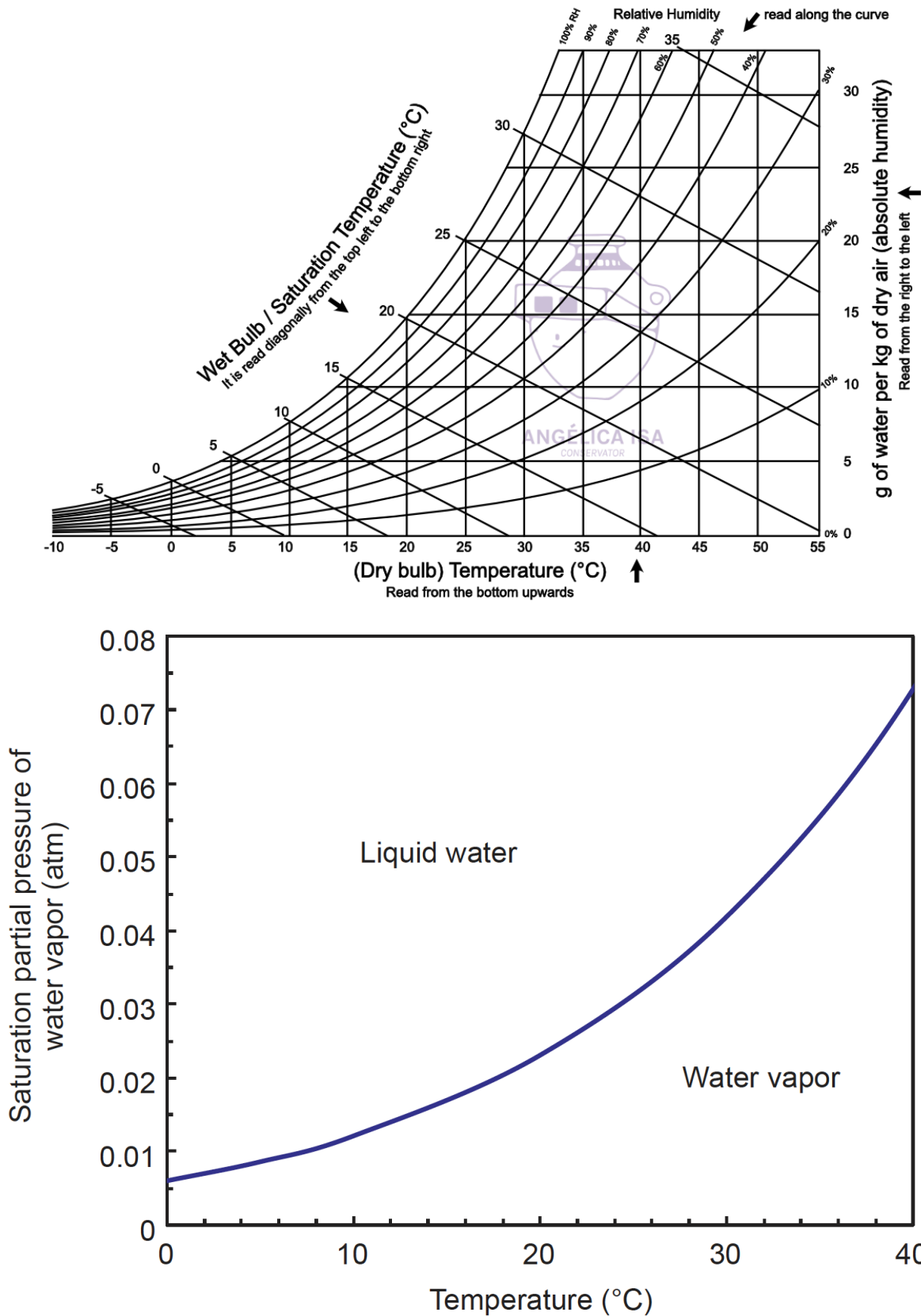


Figure 8.1: **Top:** Psychrometric chart for water at standard pressure conditions 1 atm = 101 kPa (from angelicaisa.com). **Bottom:** Phase transition coexistence curve for water near the triple point (from Mathez & Smerdon).

Record your observations here and give an explanation for what is occurring.

If you examine the full phase diagram from last week, you see that the coexistence curve between liquid water and solid water is basically vertical, meaning it has the same transition temperature regardless of the pressure value. Try to observe this **frost point** by cooling your water further (this can be accomplished by adding salt to the water, as “freezing point depression” lowers the transition temperature within your beaker). Record any observations and temperature measurements.

8.2.2 Drying your hands

You are familiar with washing your hands and letting them dry off. Let’s try to quantify that experience, slightly. Measure the temperature of “cold” water from the tap. Then run your hands under the water for a few seconds. Turn off the water and shake off your hands five times to remove large water droplets. Now wave your hands back and forth in the air in a regular motion while your partner times you with a stopwatch. Stop the timer when your palms are dry, as determined by your partner. Repeat the whole experiment with “hot” water from the tap. Record your temperatures, times, and observations here.

8.2.3 An alcohol upside-down cloud

Particle physicists use “cloud chambers” to observe the paths of particles in nuclear reactions (break up of atoms into smaller atoms by fission) and particle collision experiments. We will build such a chamber using isopropyl alcohol (isopropanol). The reason for using alcohol for such an experiment is because it actually *does not* make clouds easily, i.e., it does not nucleate as easily as other substances. A **cloud** is actually not composed of vapor, but is instead small liquid or ice droplets that have formed when vapor condenses. So, in the cloud chambers we will build (and that particle physicists use) the air is *super saturated* with alcohol and makes a “cloud” (the trail of tiny liquid particles) only when driven to nucleation by the ionization of air/alcohol by the passing particle.

Procedure:

- Construct your cloud chamber by doing the following: (1) Fold the piece of felt in half and cutting it so that the folded piece matches the size of the cubical box. (2) Attach the felt to the base of the box using magnets. (3) Cover the bottom (out) side of the baking pan with black acrylic (4) Soak the felt with alcohol; it should be wet, but if there is excess liquid in your box you should pour that out. (5) Place the radioactive rock at the center of your upside-down baking sheet (your instructor will bring it over using gloves). (6) Place the box upside-down on the baking sheet and caulk around the edges to seal it. (7) Obtain (your instructor will bring) a chunk of dry ice; place it on the corkboard mat; place the baking sheet box on top of the dry ice.
- The clouds are best observed in a mostly dark space, using a flashlight. Look for “streams” of cloud emerging from the radioactive rock

Draw your setup and describe your observations.

8.2.4 Clouds

Now we will build clouds made of water vapor. The reason the alcohol clouds were “upside down” (the alcohol was placed on the ceiling, the clouds formed down at the cold bottom layer) was because alcohol vapor is *more dense* than air and therefore sinks. Water vapor is *less dense* than water. So, in our environment outside, water evaporates off of lakes and oceans and then rises upwards. When the temperature drops sufficiently, the water vapor (invisible) becomes liquid or ice droplets and makes a visible cloud. So, in this experiment, we will have a jar with (warm) water at the base and a “roof” that is cooled a bit below 0°C.